Astronomy A2: Solutions

Light, Telescopes, and Stars · MAIS PAC

Problem 1.

(a) Neglecting the small secondary-mirror obstruction, the collecting-area ratio is $(\mathbf{D}/\mathbf{d_e})^2 = (200 / 7.0)^2 \approx \mathbf{816}$. A star at the telescope's detection threshold can therefore be 816 times dimmer than a 6.0-magnitude star:

$$\mathbf{m}_{\text{lim}} = 6.0 + 2.5 \log_{10}(816) \approx 6.0 + 7.28 \approx 13.3, \tag{1}$$

not bright enough, in this idealized estimate, for Pluto at ~14.3 mag; reaching Pluto requires a larger aperture, detector integration or other gains. The same aperture can still show many galaxies and globular clusters.

(b) At $\lambda = 5.50 \times 10^{-7}$ m and $\mathbf{D} = 0.20$ m, $\theta_{\text{min}} = 1.22(5.50 \times 10^{-7})/0.20 = 3.36 \times 10^{-6}$ rad $\approx \mathbf{0.69}''$. From the ground, turbulent temperature pockets in Earth's atmosphere ("astronomical seeing") refract starlight randomly over ~0.5″–2″, blurring large telescopes unless deformable adaptive-optics mirrors correct the wavefront in real time.

Problem 2.

(a) By Wien's displacement law, $\mathbf{T} = \mathbf{b}/\lambda_{\text{max}} = (2.90 \times 10^{-3}\ \text{m·K})/(8.30 \times 10^{-7}\ \text{m}) \approx \mathbf{3{,}490\ K}$ (≈ 3,500 K), giving Betelgeuse its unmistakable orange-red glow in winter skies.

(b) Using $(\mathbf{L})/(\mathbf{L}_\odot) = ((\mathbf{R})/(\mathbf{R}_\odot))^2\,((\mathbf{T})/(\mathbf{T}_\odot))^4$:

$$(\mathbf{R})/(\mathbf{R}_\odot) = \sqrt{((\mathbf{L})/(\mathbf{L}_\odot))}\,((\mathbf{T}_\odot)/(\mathbf{T}))^2 \approx \sqrt{(10^5)}\,((5{,}780)/(3{,}490))^2 \approx 316 \times 2.74 \approx 870\ \mathbf{R}_\odot. \tag{2}$$

Dividing by 215 $\mathbf{R}_\odot$/AU gives $\mathbf{R} \approx \mathbf{4.0\ AU}$—if placed at the center of our Solar System, Betelgeuse would engulf Mercury, Venus, Earth, Mars, and much of the asteroid belt!

Problem 3.

(a) At orbital distance $\mathbf{a}$, stellar flux is $\mathbf{F} = \mathbf{L}_\star/(4\pi\mathbf{a}^2) = (4\pi\mathbf{R}_\star{}^2\sigma\mathbf{T}_\star{}^4)/(4\pi\mathbf{a}^2) = \sigma\mathbf{T}_\star{}^4(\mathbf{R}_\star/\mathbf{a})^2$. Equating absorbed power $\pi\mathbf{R_p}{}^2(1-\mathbf{A})\mathbf{F}$ to thermally emitted power $4\pi\mathbf{R_p}{}^2\sigma\mathbf{T}_{\text{eq}}{}^4$ cancels $\pi\mathbf{R_p}{}^2\sigma$ on both sides:

$$4\,\mathbf{T}_{\text{eq}}{}^4 = (1-\mathbf{A})\,\mathbf{T}_\star{}^4\,((\mathbf{R}_\star)/(\mathbf{a}))^2 \Rightarrow \mathbf{T}_{\text{eq}} = \mathbf{T}_\star\,(1-\mathbf{A})^{1/4}\,\sqrt{((\mathbf{R}_\star)/(2\mathbf{a}))}. \tag{3}$$

(b) Substituting Earth's numbers gives $\mathbf{T}_{\text{eq}} = 5{,}780 \times (0.70)^{1/4} \times \sqrt{(6.96 \times 10^8 / 3.00 \times 10^{11})} \approx 5{,}780 \times 0.9147 \times 0.04817 \approx \mathbf{255\ K\ (-18\ °C)}$. Without an atmosphere, Earth's oceans would freeze solid; trace greenhouse gases (water vapor and CO_2) are transparent to incoming visible sunlight but absorb outgoing thermal infrared radiation, warming the surface

by ~33 K to a habitable 288 K (+15 °C).

Problem 4.

(a) Available fusion energy is $\mathbf{E}_{\text{tot}} \approx 0.007 \times (0.10\ \mathbf{M}_{\odot})\mathbf{c}^2 = 7.0 \times 10^{-4}\ (1.99 \times 10^{30}\ \text{kg})(3.00 \times 10^{8}\ \text{m/s})^2 \approx 1.25 \times 10^{44}$ J. Dividing by $\mathbf{L}_{\odot} = 3.83 \times 10^{26}$ W yields

$$\tau_{\odot} \approx (1.25 \times 10^{44}\ \text{J})/\ (3.83 \times 10^{26}\ \text{W}) \approx 3.27 \times 10^{17}\ \text{s} \approx 10.4\ \text{Gyr} \tag{4}$$

(comfortably longer than the Solar System's current 4.6 Gyr age).

(b) Since fuel reserves scale as $\mathbf{M}$ while burn rate scales as $\mathbf{L} \propto \mathbf{M}^{3.5}$, lifetime scales inversely with mass:

$$\tau_{\text{MS}} \propto (\mathbf{M})/\ (\mathbf{L}) \propto \mathbf{M}^{-2.5} \Rightarrow \tau_{\text{MS}} \approx 10\ \text{Gyr}\ ((\mathbf{M})/\ (\mathbf{M}_{\odot}))^{-2.5}. \tag{5}$$

A massive 10 $\mathbf{M}_{\odot}$ star shines ~3,160 times brighter than the Sun and exhausts its core in just 10 Gyr $\times\ 10^{-2.5} \approx$ **32 million years**, exploding as a core-collapse supernova. Conversely, a thrifty 0.20 $\mathbf{M}_{\odot}$ red dwarf lasts 10 Gyr $\times\ (0.20)^{-2.5} \approx$ **560 Gyr** (this is only the stated crude scaling extrapolated to low mass; real low-mass stars have a different mass–luminosity slope and can mix a larger fuel fraction, so detailed lifetimes differ)—more than 40 times the present 13.8 Gyr age of the universe!